

Differential: M1 supplement

PRV 201 Frontiers of Technology (AI) · Track A: Build · Stevens Institute of Technology · September 2026

This supplement accompanies the M1 problem statement. The statement stands on its own; nothing here is needed to read it.

Industry watch, September 2026

- **Frontier models score highly on textbook analog questions.** The top Razavi-Bench model scored 95.50%, graded by another language model, and the benchmark has an experimental simulator-assisted mode [1]. A bench tool must add what textbooks do not test: tolerances, ambiguity between faults, the cost of each measurement and safety. T5 compares the full system with such a model working alone.
- **IBM Maximo 9.2 (June 2026)** adds agentic workflows and an MCP server through which customers can connect their own agents [2], a possible route for a diagnosis engine into industrial repair depots.
- **EU AI Act transparency rules apply from 2 August 2026** [3]; the 2026 amendments moved the Annex III high-risk rules to 2 December 2027 [4] and keep a duty on providers and deployers to support their staff's AI literacy [5]. They would bind Differential only once offered in the EU.
- **Colorado re-enacted its AI law** as a regime for automated decision-making technology in consequential decisions, approved on 14 May 2026 and in force from 1 January 2027 [6]. A shop that used repair tickets to decide about its staff could come within such rules; Differential's license forbids that use.
- **Claude Sonnet 5.5 was released on 28 September 2026** [7]. Models change quickly, so Differential pins its model, caches responses and re-runs its safety and grounding targets (T15-T19) on any change.

Course coverage map

Table 1. Course topics and where the project uses them, with parts named by title.

Wk	Topic	Where the project uses it	Depth
2	AI/ML algorithms	Bayesian inference, Gaussian mixtures, boosted trees, information-gain measurement choice (Part 4 — Preliminary AI approach, risks and ethics)	Core
3	Python	The simulation, engine and evaluation code that produces every number in the statement (repository); no code appears in the statement itself	Core (code)
4	Large language models	Chat-model benchmark scores (Part 1 — Problem, stakeholders and industry context); the language-model baselines (Part 3 — Objectives and measurable success criteria; T5); why language models guess (Part 4 — Preliminary AI approach, risks and ethics)	Core
5	LLM-enhanced agents	An assistant whose tools act only on the diagnosis session; complaint reading and explanation (Part 4 — Preliminary AI approach, risks and ethics)	Supporting
6	Building ML systems	Pilots, locked targets with confidence bounds, stress sets, hardware protocol (Part 3 — Objectives and measurable success criteria)	Core
7	NLP	Complaint reading with noise; sealed paraphrase bank (Part 3 — Objectives and measurable success criteria; T18)	Supporting
8	Image recognition	Meter-photo reading scored on value, unit and mode (T19)	Supporting
9	Facial recognition (bias auditing)	Error breakdown by who bears the errors: complaint writers, technicians, owners of undocumented gear (Part 4 — Preliminary AI approach, risks and ethics)	Supporting
10	Autonomous transportation	Five levels of diagnostic autonomy, L0 (the technician does everything) to L4 (an unattended fixture), defined by analogy to driving automation; Differential placed at L2 on safety and accountability grounds (Part 4 — Preliminary AI approach, risks and ethics)	Supporting
11	Robotics	Not covered. Reading bench instruments over SCPI is test automation, not robotics	None
12	Cybersecurity	Prompt injection, untrusted data, session-only tools, adversarial prompts (Part 4 — Preliminary AI approach, risks and ethics; T16)	Core
13	Ethics and governance	Hazard audit, OSHA, liability, EU AI Act and product liability, research ethics, NIST AI RMF (Part 1 — Problem, stakeholders and industry context, Part 4 — Preliminary AI approach, risks and ethics)	Core

References

- [1] Zhishuai Zhang, “Razavi-Bench: An Expert-Curated Benchmark for Analog-Design Reasoning,” *Benchmark repository and leaderboard (GitHub Arcadia-1/razavi-bench)*, Sep. 2026. [Online]. Available: <https://github.com/Arcadia-1/razavi-bench> (accessed Sep. 28, 2026).
- [2] IBM, “Introducing Maximo Application Suite 9.2: AI built for asset management that keeps operations running,” Press release, Jun. 25, 2026. [Online]. Available: <https://www.ibm.com/new/announcements/introducing-maximo-application-suite-9-2> (accessed Sep. 28, 2026).
- [3] European Commission, “Commission starts enforcing AI Act rules and new transparency requirements on 2 August (IP/26/1714),” Press release, Jul. 31, 2026. [Online]. Available: <https://digital-strategy.ec.europa.eu/en/news/commission-starts-enforcing-ai-act-rules-and-new-transparency-requirements-2-august> (accessed Sep. 28, 2026).
- [4] European Parliament and Council of the European Union, “Regulation (EU) 2026/1744 of 8 July 2026 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI),” *Official Journal of the European Union, OJ L*, 2026/1744, Jul. 24, 2026. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2026/1744/oj/eng> (accessed Sep. 28, 2026).
- [5] Publications Office of the European Union, “Regulation (EU) 2024/1689, consolidated text 02024R1689-20260727 (incorporating Regulation (EU) 2026/1744),” *EUR-Lex*, Jul. 27, 2026. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2024/1689/2026-07-27/eng> (accessed Sep. 28, 2026).
- [6] Colorado General Assembly, “SB26-189, Automated Decision-Making Technology,” May 14, 2026. [Online]. Available: <https://leg.colorado.gov/bills/sb26-189> (accessed Sep. 28, 2026).
- [7] Anthropic, “Claude Sonnet 5.5 (model page),” *Claude Developer Platform documentation*, Sep. 28, 2026. [Online]. Available: <https://platform.claude.com/docs/en/models/sonnet-5-5/overview> (accessed Sep. 28, 2026).